

How Machine Learning Works

Input1	Input2	Input3	Output
1	1	-1	1
2	-1	-2	-1
1	-1	0	0
-1	-1	3	1

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$$Output = Input1 + Input2 + Input3$$

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Training

$$\text{Output} = \text{Input1} + \text{Input2} + \text{Input3}$$

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Training

$$\text{Output} = \text{Input1} + \text{Input2} + \text{Input3}$$



$$f(x_1, x_2, x_3) = x_1 + x_2 + x_3$$

Model

How Machine Learning Works

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2	-1	-2	-1
1	-1	0	0
-1	-1	3	1
3	2	-5	?

Training

Inference

$$\text{Output} = \text{Input1} + \text{Input2} + \text{Input3}$$



$$f(x_1, x_2, x_3) = x_1 + x_2 + x_3$$

Model

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Training

Inference

$$\text{Output} = \text{Input1} + \text{Input2} + \text{Input3}$$



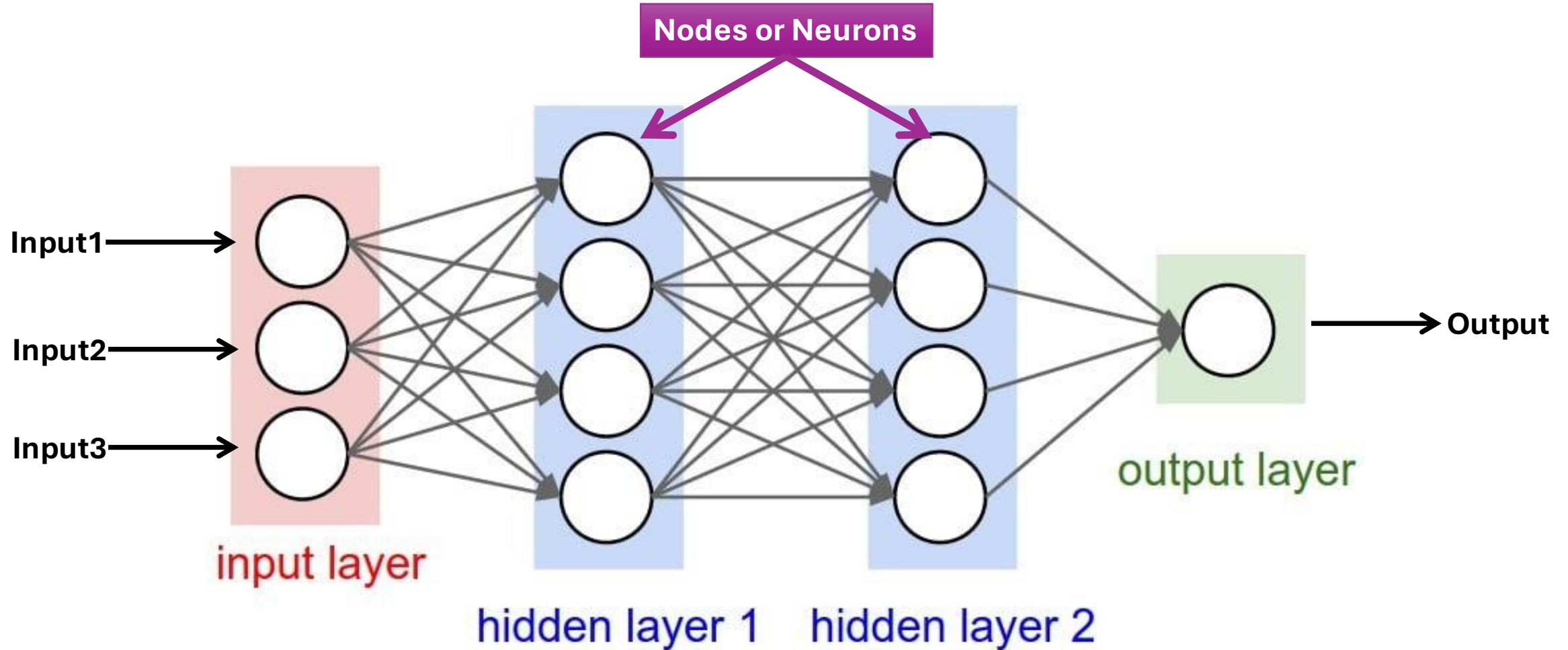
$$f(x_1, x_2, x_3) = x_1 + x_2 + x_3$$

Model

Neural Network Architecture

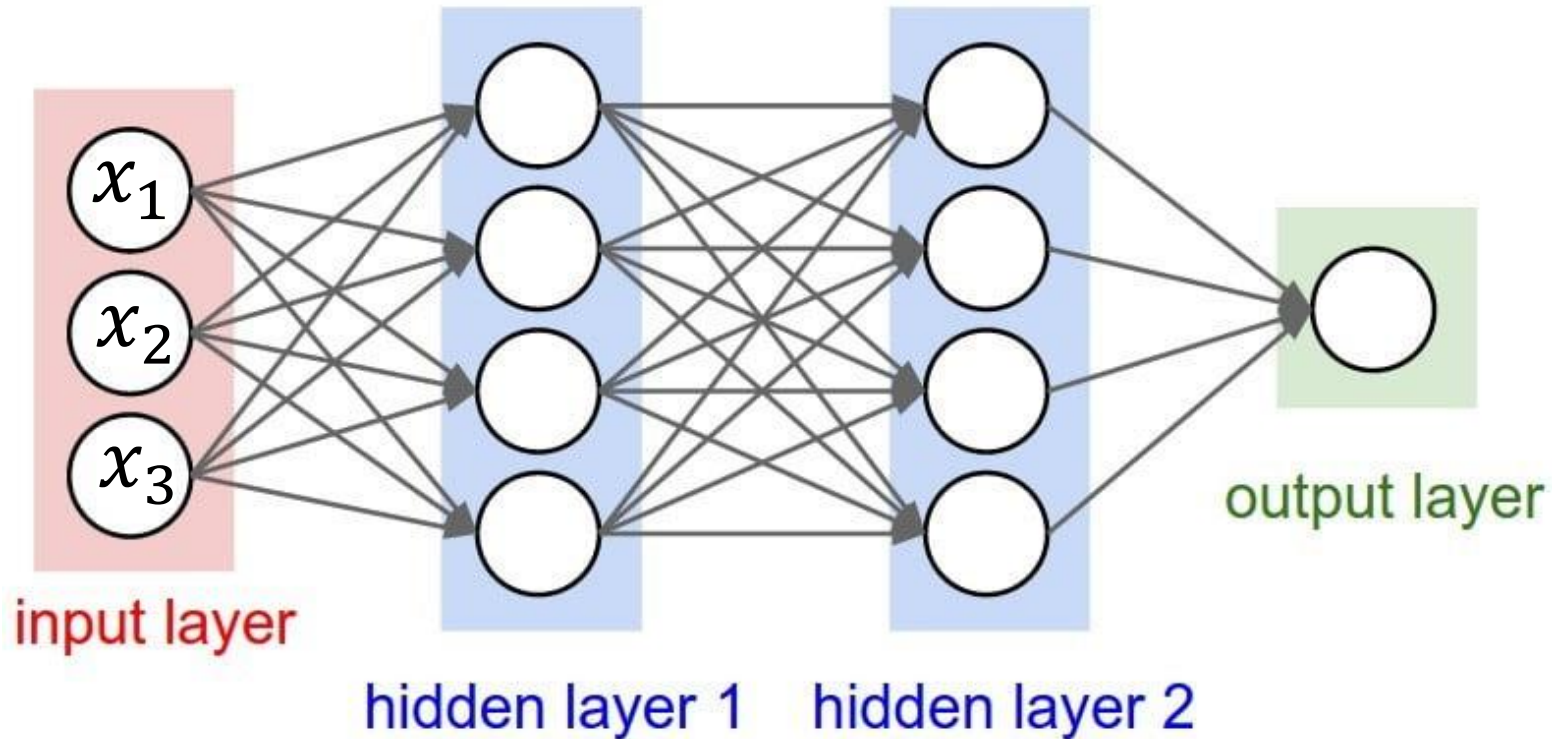
Neural Network Architecture

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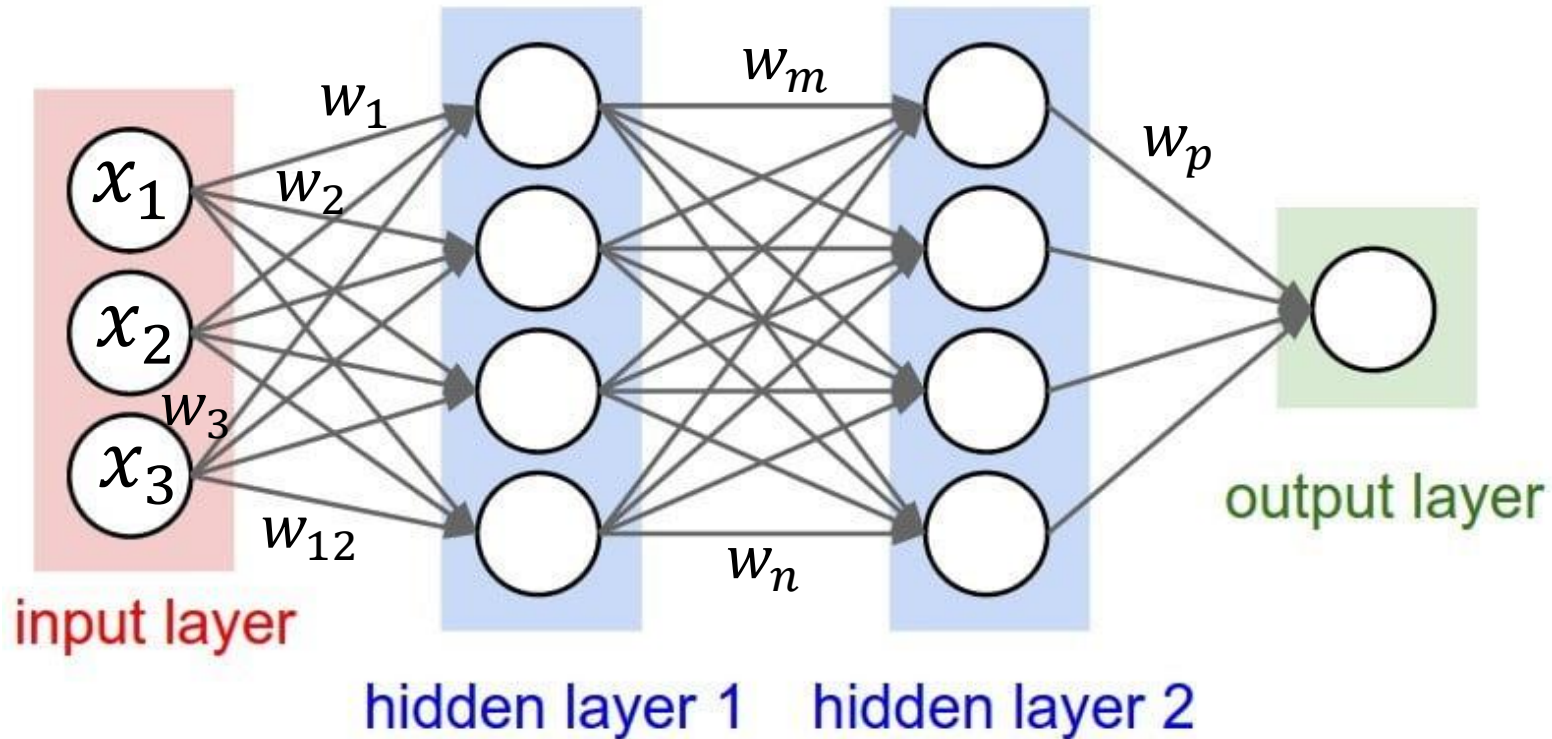
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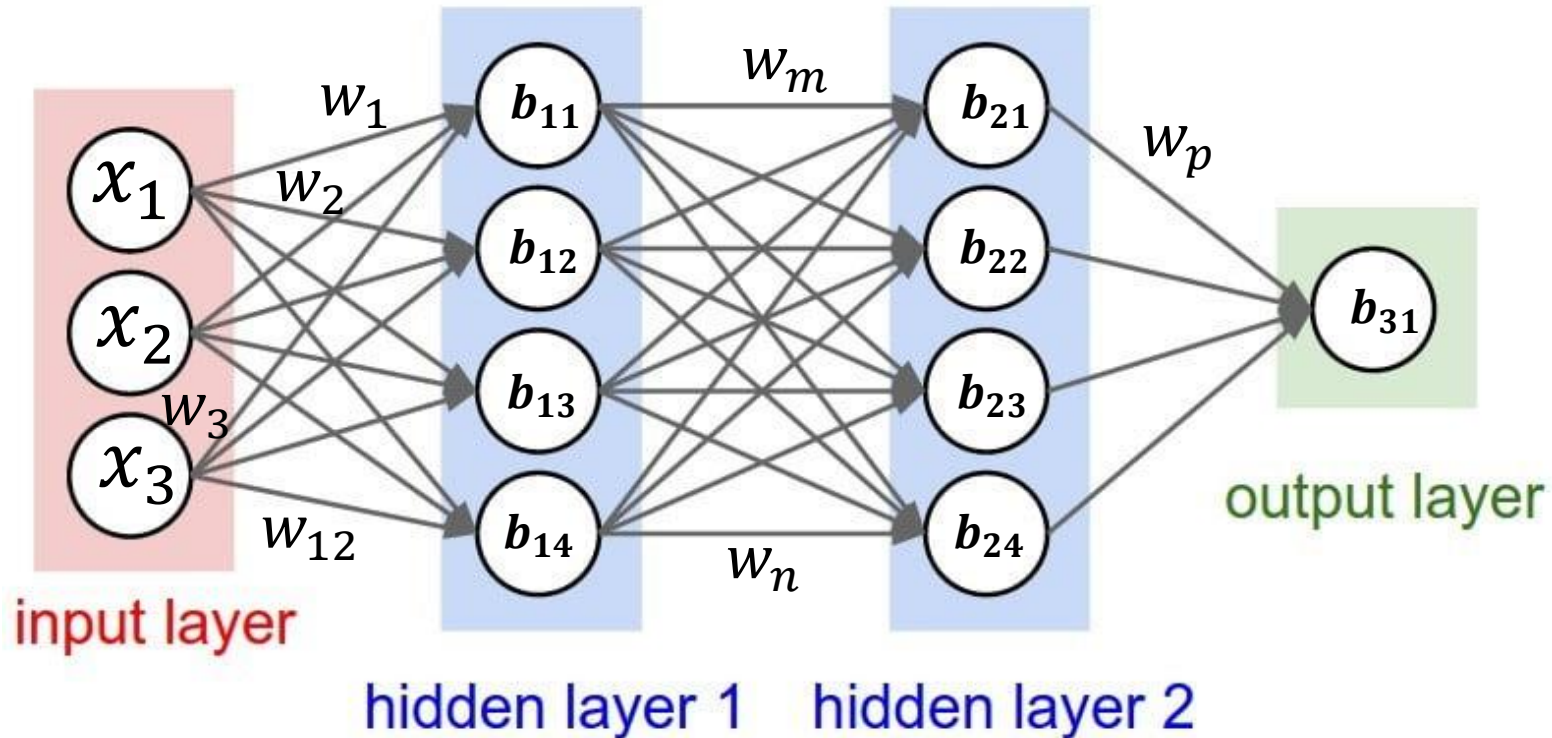
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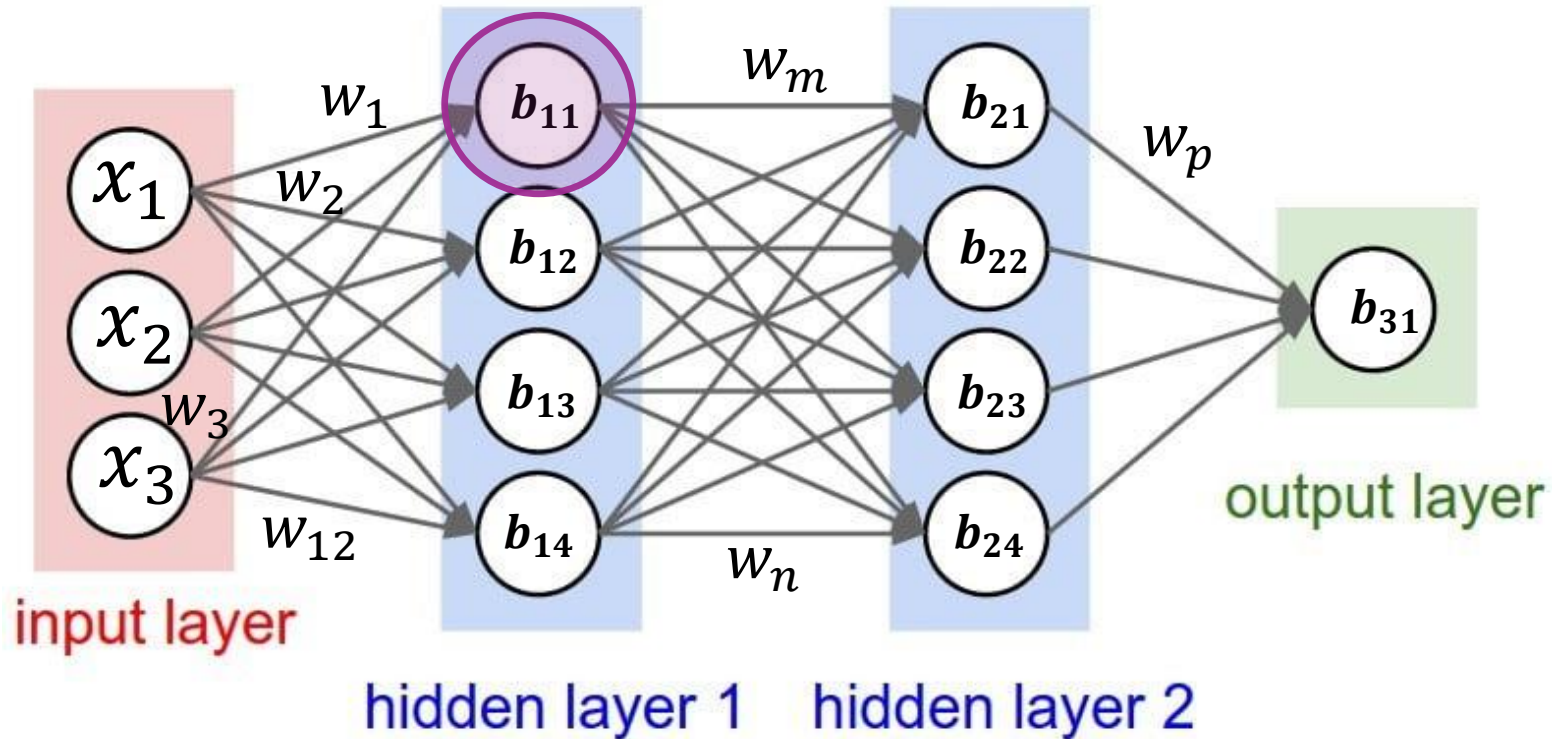
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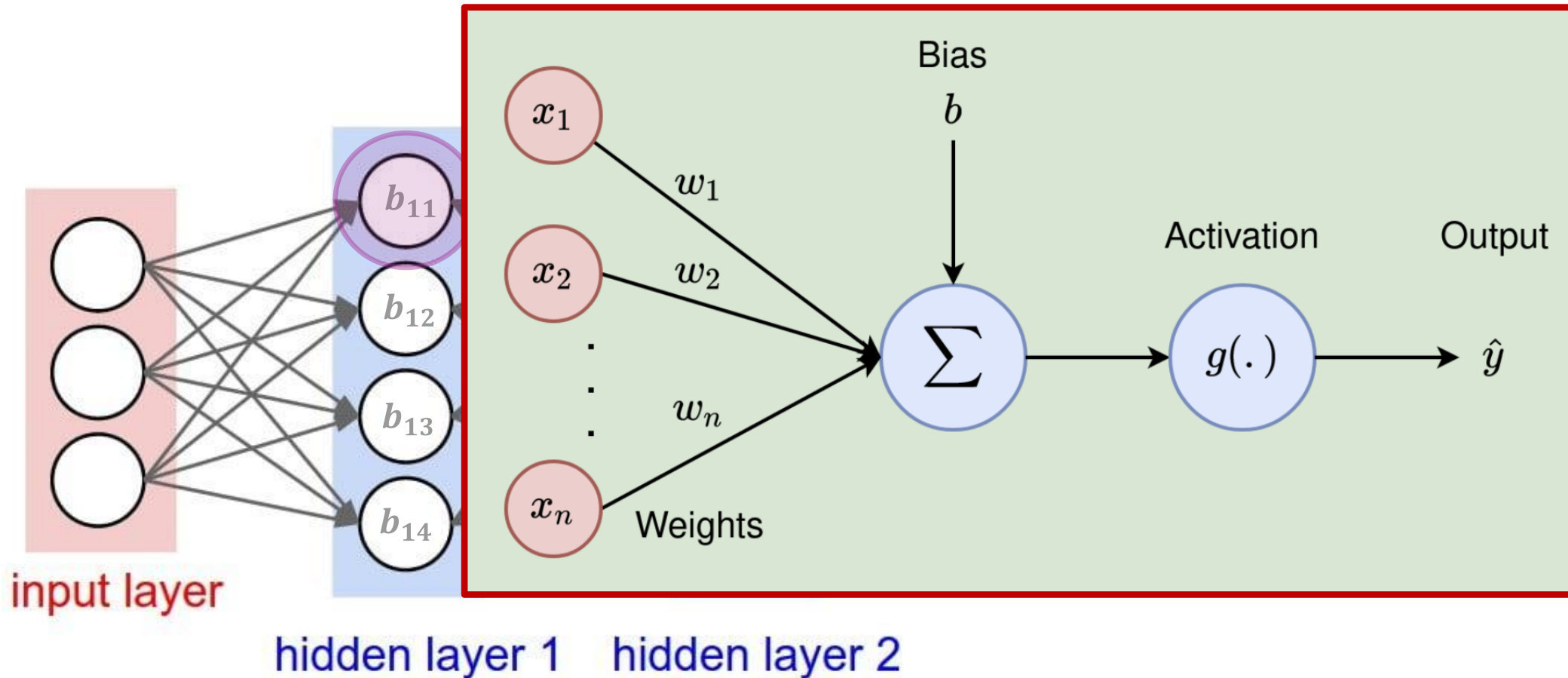
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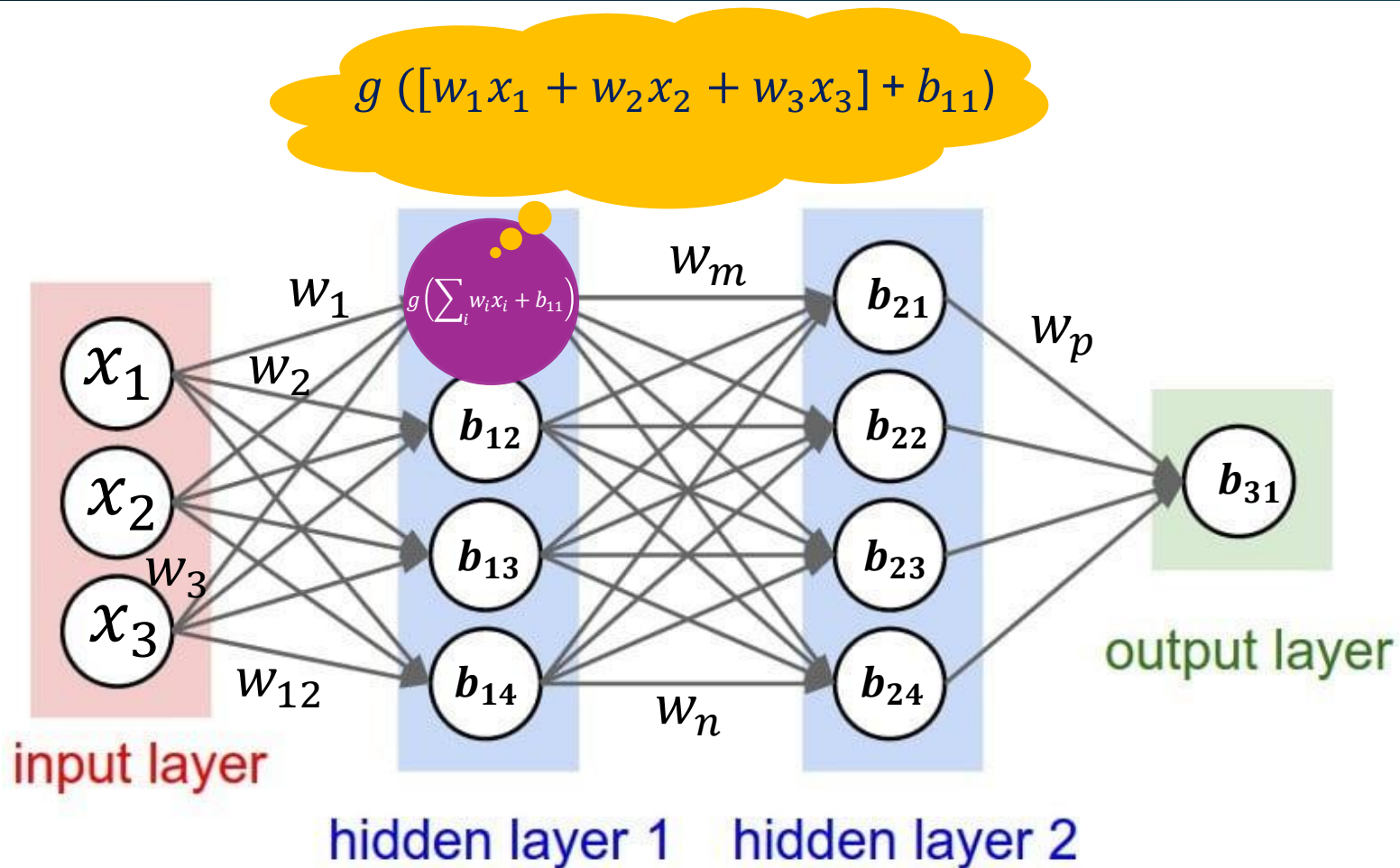
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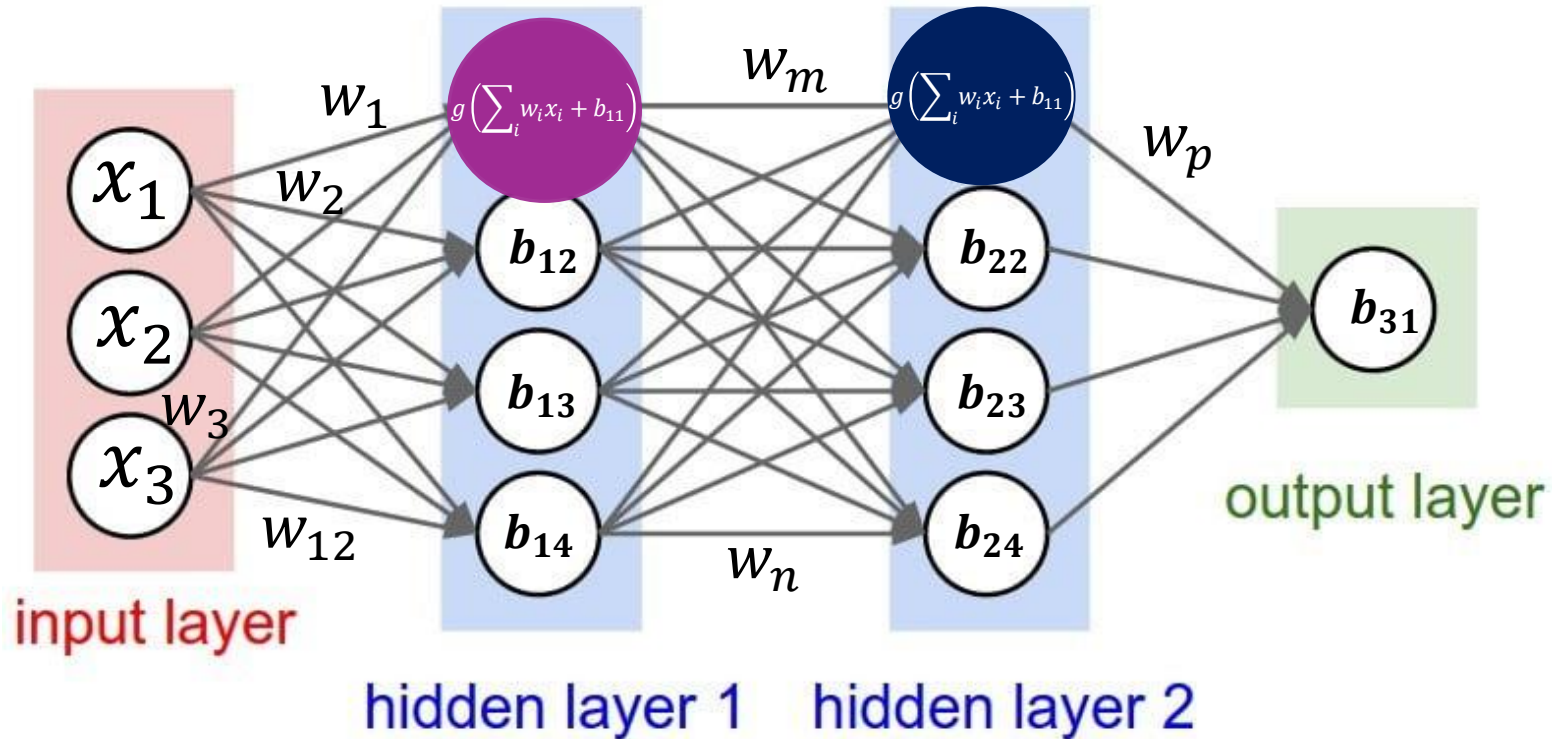
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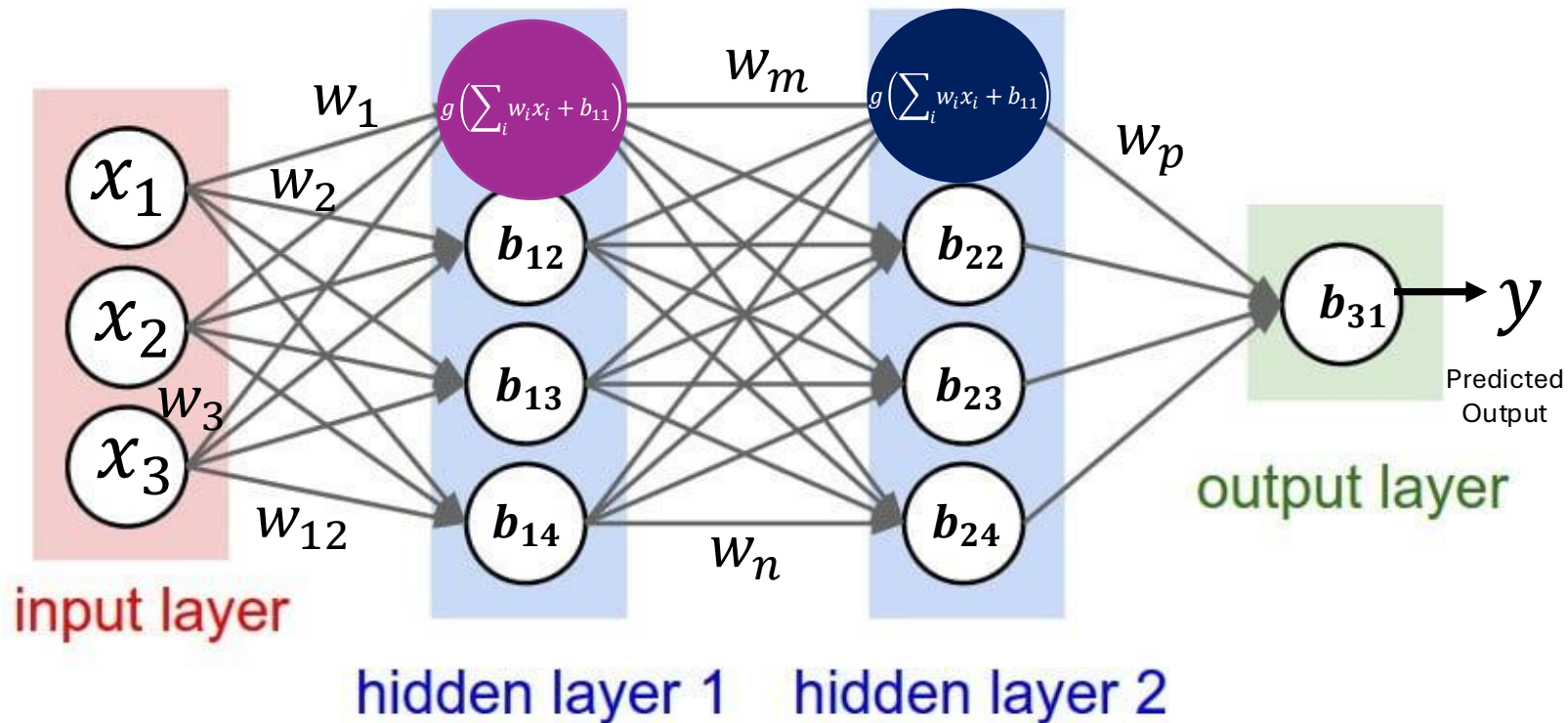
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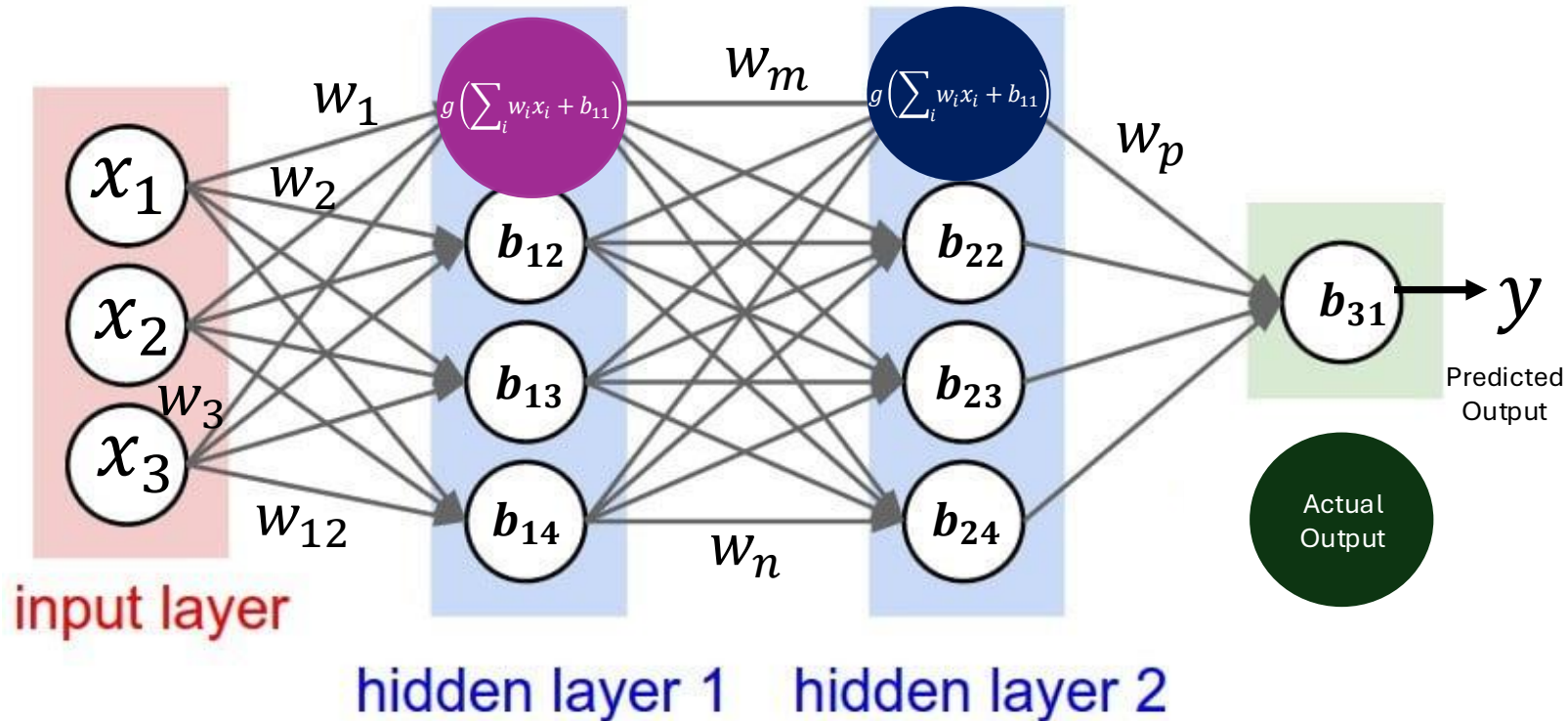
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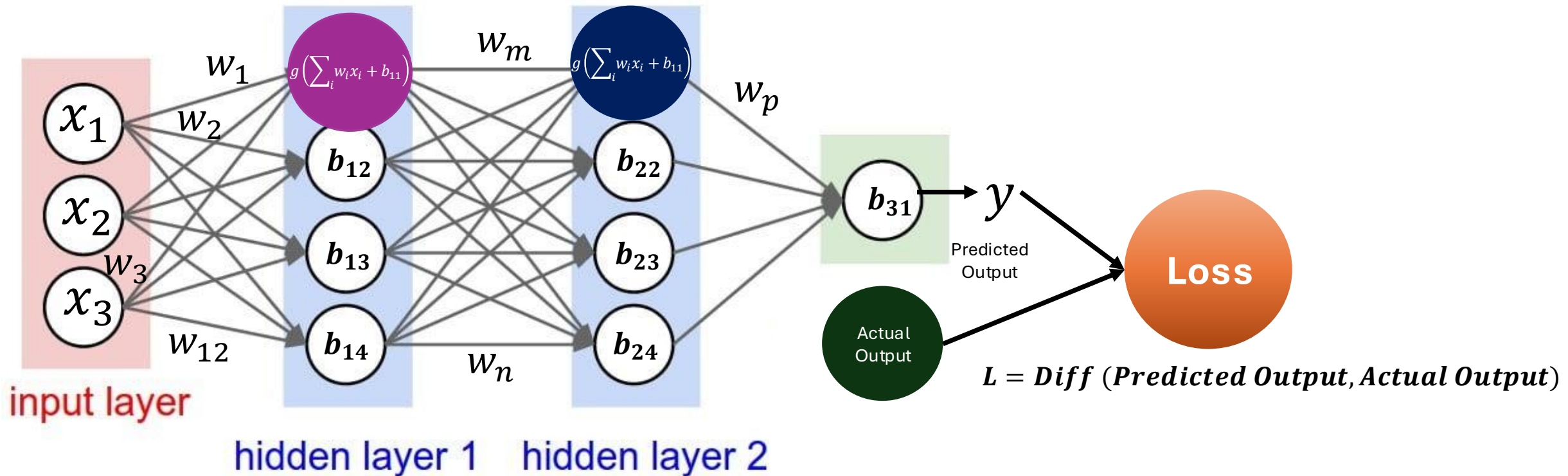
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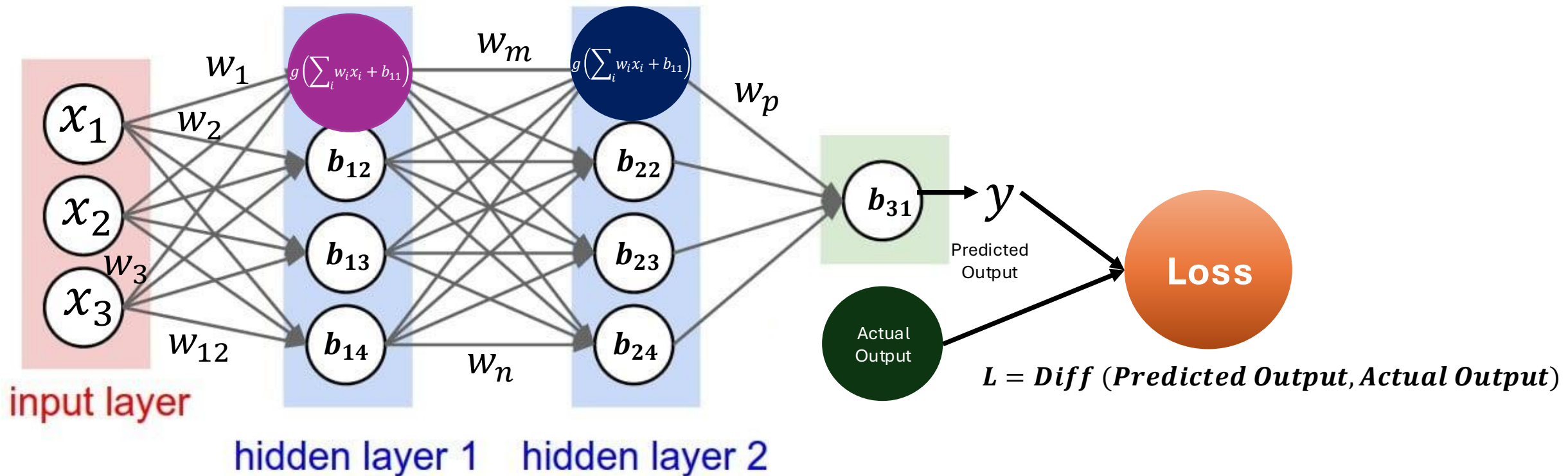
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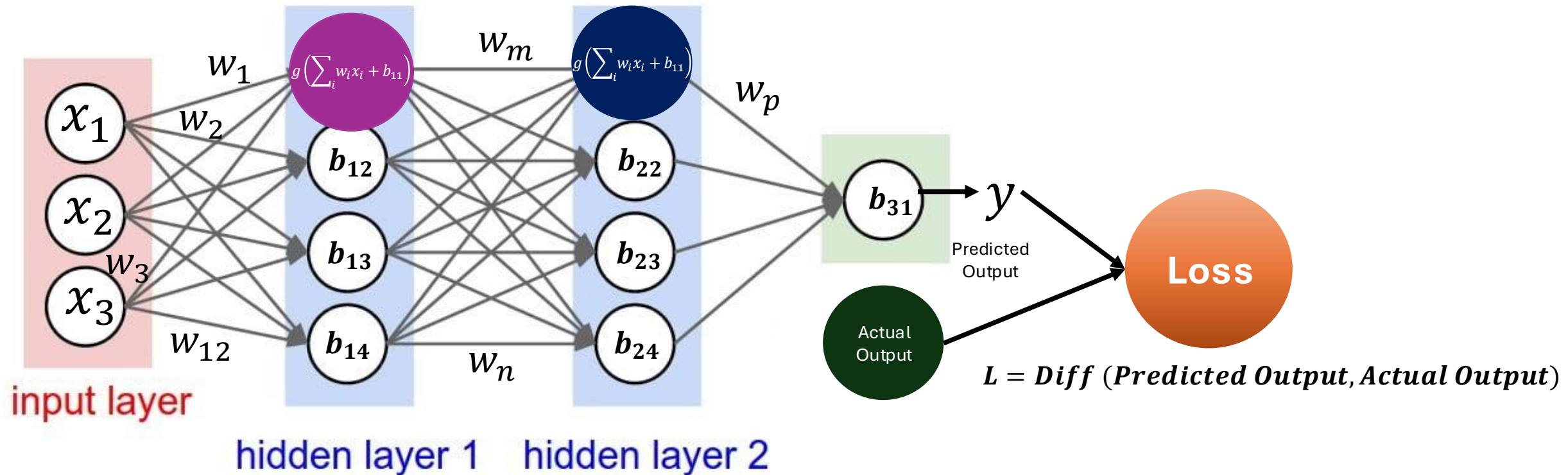
$$L = \text{Diff}(\text{Predicted Output}, \text{Actual Output})$$

Step-1: Forward Propagation

Neural Network Architecture

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Step-2: Backward Propagation

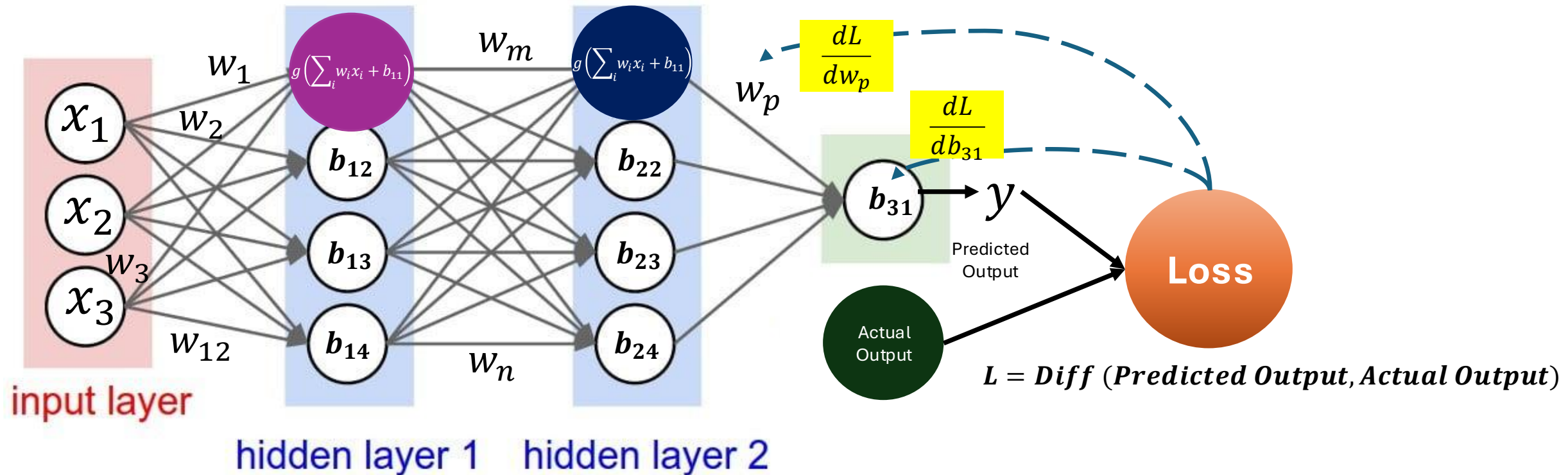


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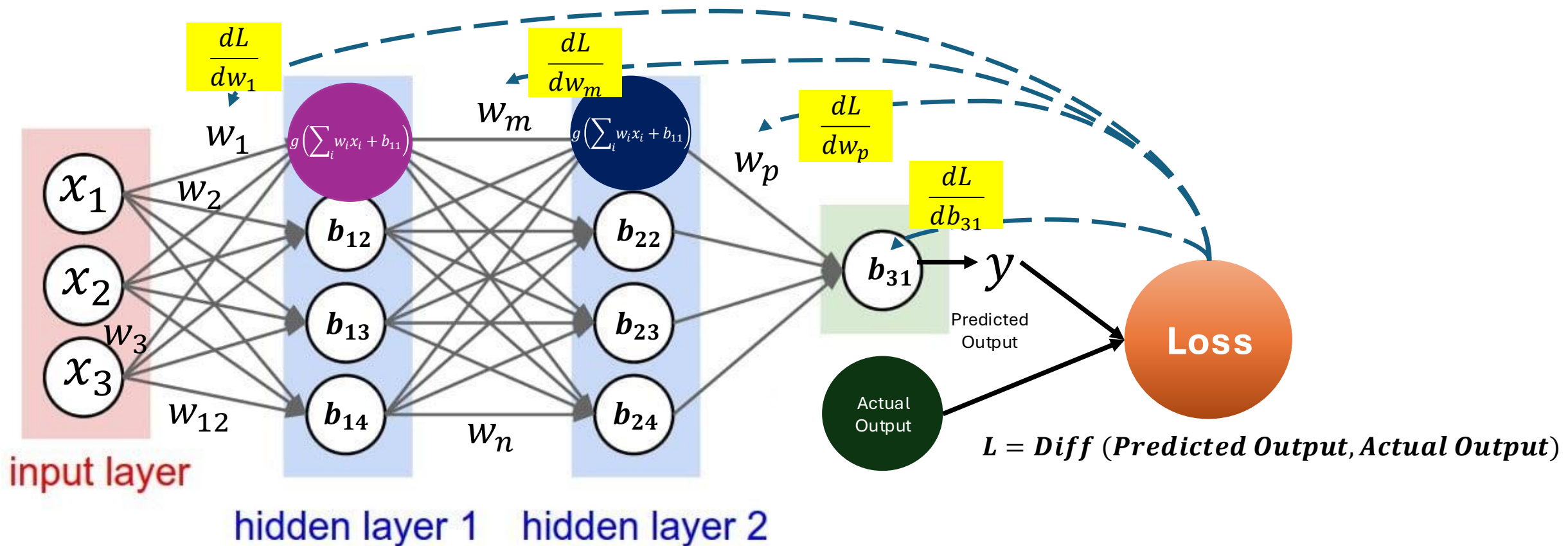


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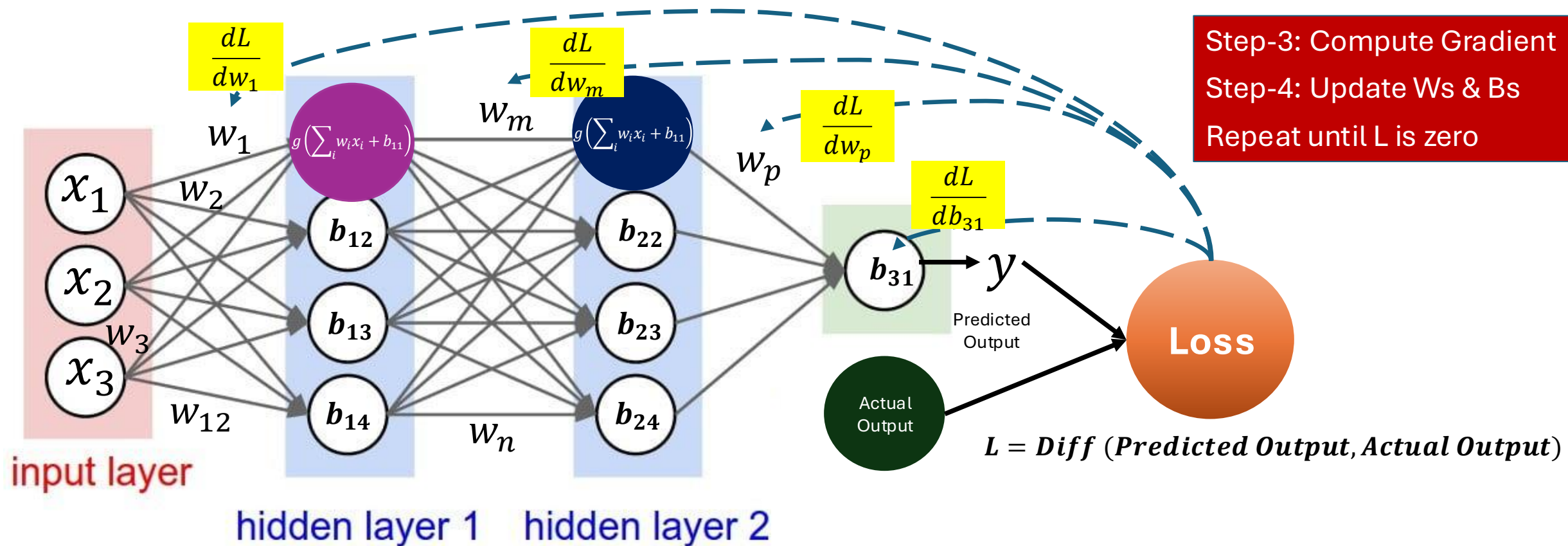


Step-1: Forward Propagation

Neural Network Architecture

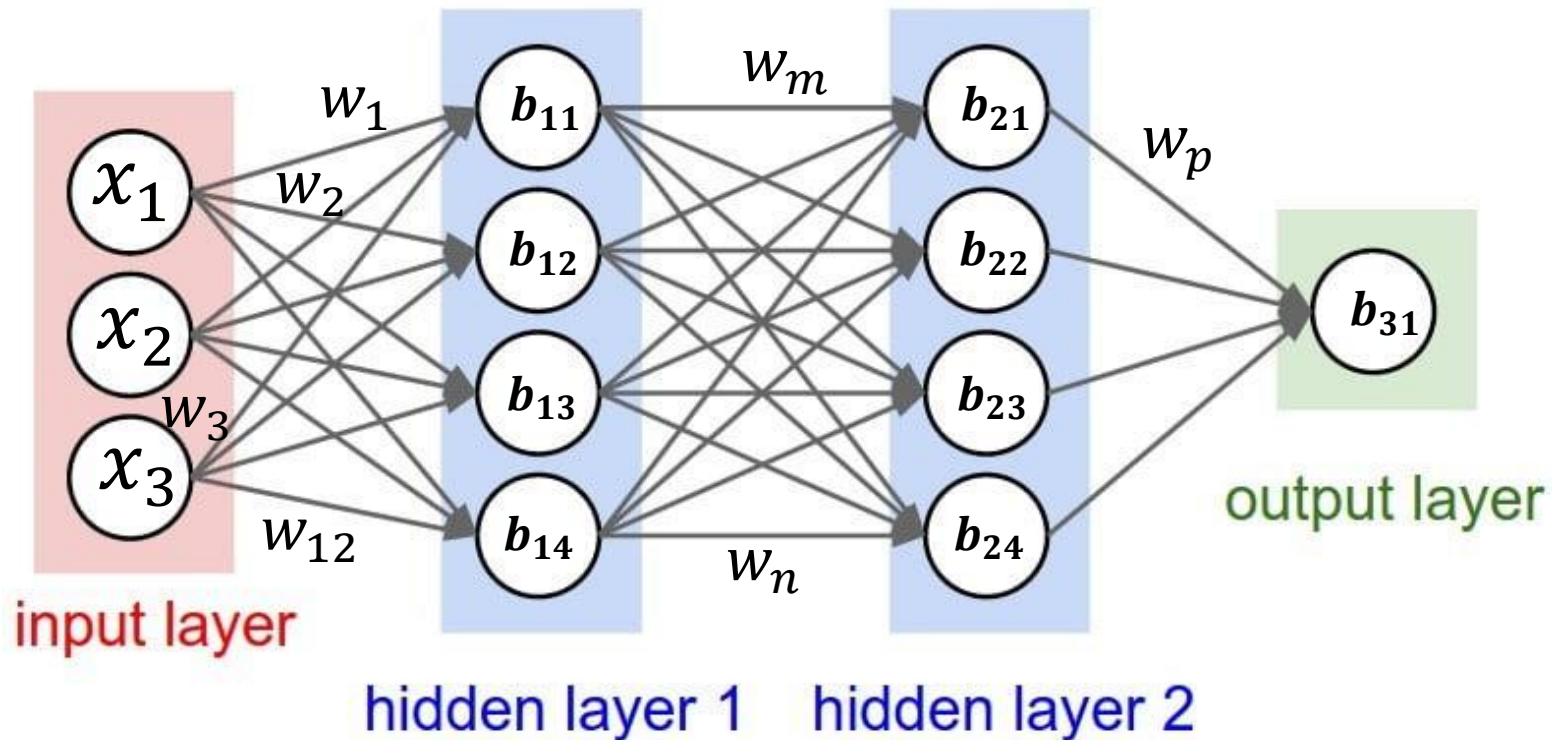
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Step-2: Backward Propagation



Step-1: Forward Propagation

Neural Network Model



At the end of the training process what we have –

- The Network, the connections between nodes, layers, etc.
- The Weights & Biases, a.k.a Parameters

Fine Tuning

Fine Tuning involves -

- Changing or Updating parameters

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- Changing or Updating parameters

Finetuning -

- Supervised Finetuning (SFT)
- Reinforcement Learning from Human Feedback (RLHF)

Fine Tuning

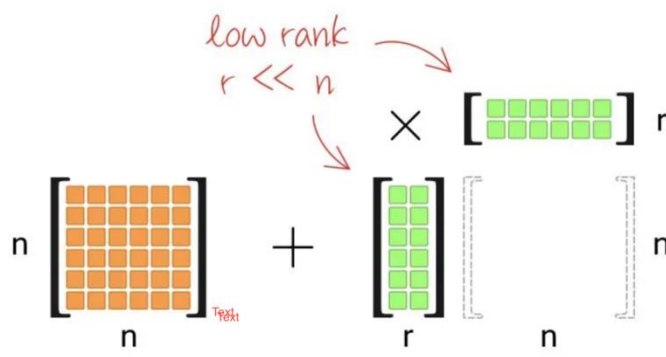
Supervised Fine Tuning -

- Full Finetuning
- Parameter-Efficient Fine-Tuning (PEFT)

Parameter-Efficient Fine-Tuning -

- LoRA (Low-Rank Adaptation)
- QLoRA (Quantized LoRA)

LoRA



$$W = W_0 + \underline{A} \underline{B}$$

low rank $r \ll n$

frozen

trainable

$$W_0 = [100K \times 100k]$$

$$A = [100K \times 1]$$

$$B = [1 \times 100K]$$

$$W = W_0 + AB$$